ICT702 Data Visualisation

Selecting a Chart Type

Module 2

I would like to acknowledge the traditional custodians of the land we are meeting on,

The Gubbi Gubbi/Kabi Kabi peoples,

and pay my respect to Elders past, present and emerging.

Part 3

Guidelines for Selecting a Chart

Chart	Relationship	Distribution	Composition	Ranking
Scatter	✓	✓		
Bubble	✓	✓		
Line	✓			
Stock	✓			
Column	✓	✓		✓
Bar	✓	✓	✓	✓
Heat Map	✓			
Choropleth Map		✓		
Stacked Bar			✓	
Stacked Column			✓	
Treemap			✓	
Waterfall			✓	
Funnel				✓

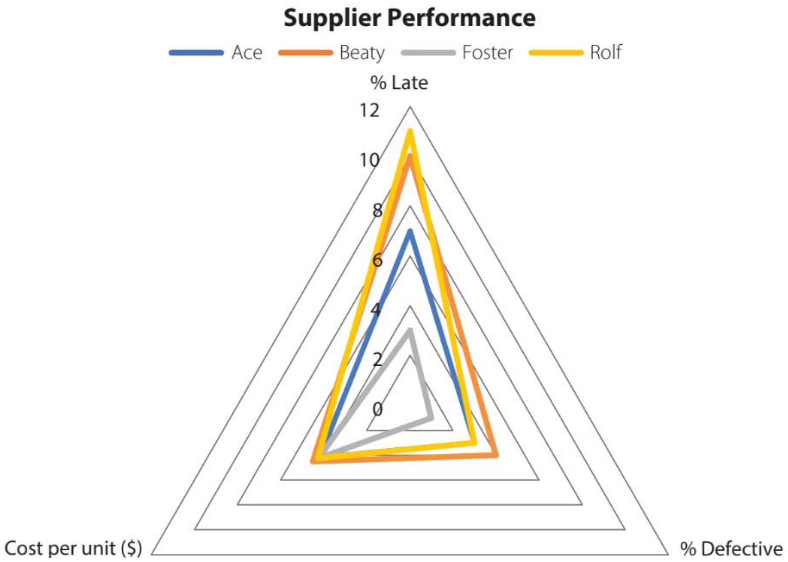
Charts to Avoid

- Pie charts with more than a few categories – bar chart is better
  - Humans are better at determining differences in length than angle and area.
  - Use of many colours increases cognitive load – bar chart colour not needed.
  - Lengthy category labels easier to read in a horizontal bar chart
- Combo charts
- 3D representations of a 2D chart.

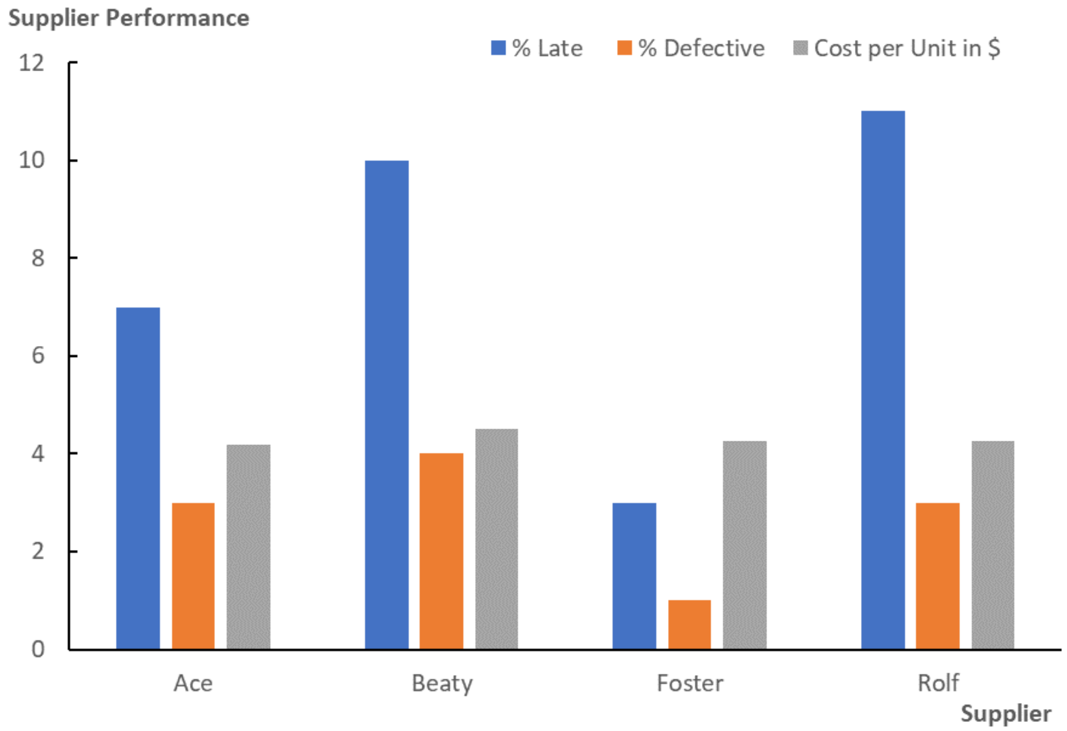
Charts to Avoid – Radar Chart

A radar chart of supplier performance for a component for Newton Industries

	A	B	C	D
1	Supplier	% Late	% Defective	Cost per Unit in $
2	Ace	7	3	$4.19
3	Beaty	10	4	$4.50
4	Foster	3	1	$4.25
5	Rolf	11	3	$4.26

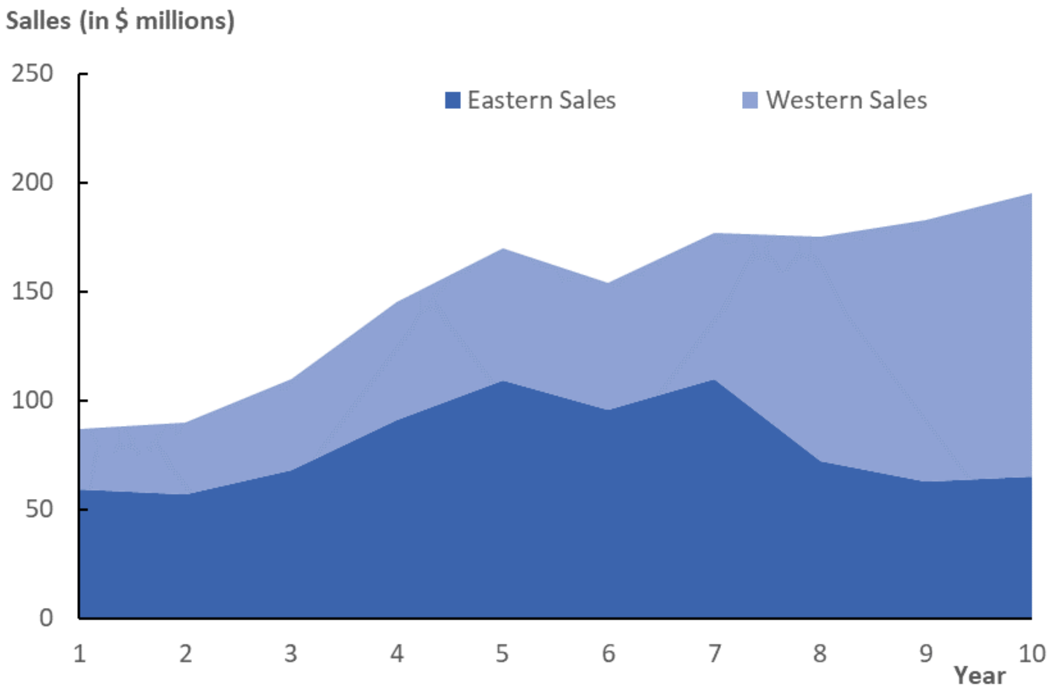


Better choice: a clustered column chart of supplier performance for a component for Newton Industries

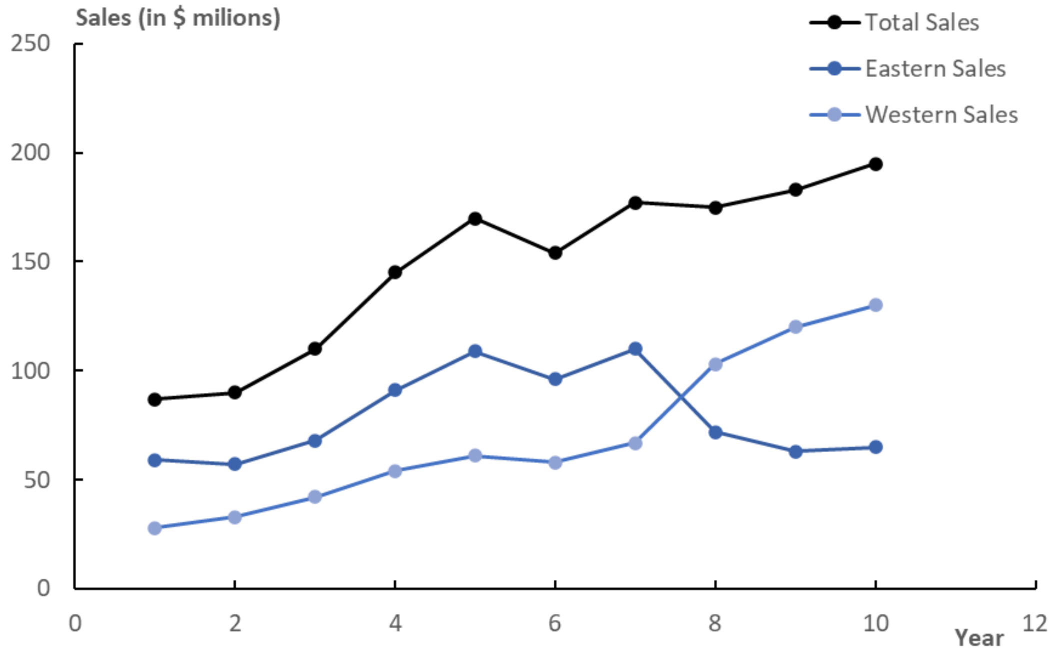


Charts to Avoid – Area Chart

An area chart for the Cheetah Sports regional sales data

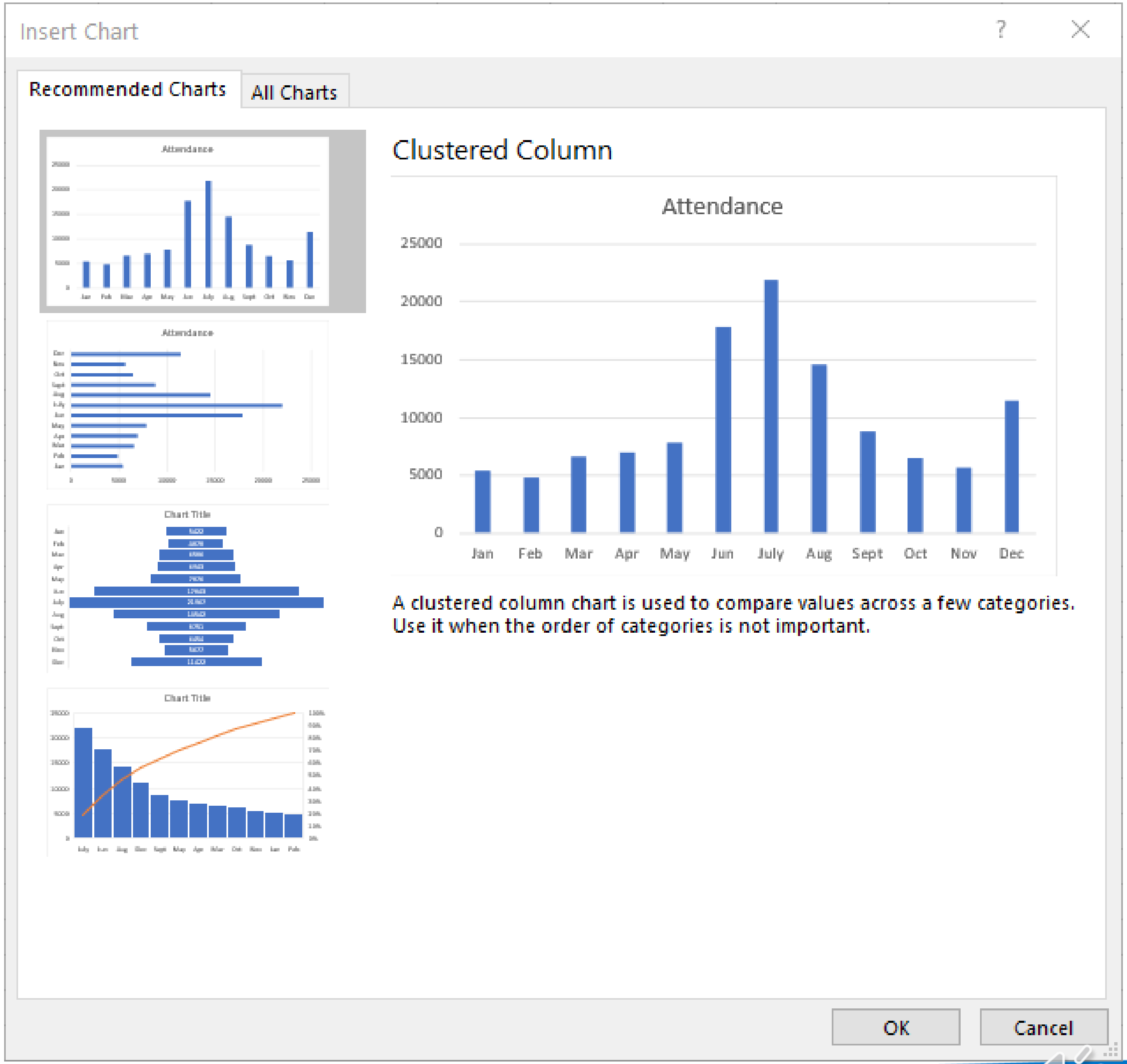


Better choice: a multiple line chart for the Cheetah Sports regional sales data



Excel's Recommended Charts Tool

The Insert Chart task pane for the Zoo Attendance data



Summary

In this module you have learned:

- to identify a suitable chart (for specific analytical goals and data types).
- These chart types include:
  - Relationship charts, such as the scatter chart, bubble chart, and line chart.
  - Composition charts, such as clustered/stacked column and bar charts.
  - Maps, such as choropleth geographical maps, heat maps, and treemaps.
  - Specialized charts, such as waterfall, stock, and funnel charts.
  - Charts to avoid, such as radar and area charts.

End of Part 3

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